

Repressive Democracies: the Relationship Between State's Repression Severity, Protests, and Policy Change.

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Abstract

This paper models the choice of repression severity in democracies as a global game of policy change. Citizens decide whether to protest without knowing how large a protest must be to force a policy change, and each observes a noisy private signal of that threshold. The state fixes severity before mobilization begins and cannot revise it during a protest. The citizens' stage has a unique equilibrium in threshold strategies, so the comparative statics hold everywhere in the parameter space without a selection assumption. The central result runs against intuition. Severity rises with the marginal cost of policing an additional protester, because severity deters participation and shrinks the crowd the state has to police. Public opinion bounds that response, since severity falls as the marginal cost of backlash rises. Severity does not respond at all to the size of the prize citizens seek. An extension makes protest motivation respond to severity. Repression backfires when the elasticity of motivation with respect to severity exceeds one, and an accelerating outrage response caps optimal severity at that point.

1 Introduction.

Scholarly work has revealed puzzling results about state repression's consequences for mobilization (Davenport 2007a). On the one hand, some studies show that threats of state repression may

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deter further mobilization by increasing its cost to protesters (Chenoweth, Perkoski, and Kang 2017; della Porta 1995; Christopher M Sullivan 2016). In opposition, others reveal that repression backfires, meaning that protesters who are victims of violence become radicalized (Adam-Troian, Çelebi, and Mahfud 2020; Christopher Michael Sullivan and Davenport 2017). Repression and dissent are strategically interlinked once severity becomes a policy variable that states choose (Chenoweth, Perkoski, and Kang 2017; Ritter 2014).

Scholars call this phenomenon the “protest-repression” nexus (Davenport 2007a; Curtice and Behlendorf 2021), and they have studied it primarily in autocracies. An empirical strand does examine protest in democracies (Carey 2006; Nam 2007; Machado, Scartascini, and Tommasi 2011; Su 2015; Aytaç, Schiumerini, and Stokes 2017), but it is largely institutional, asking what makes protest more or less likely across democratic settings rather than how a state chooses the severity with which it meets protest. This is especially true when theoretical or micro-founded approaches have been put forth to explain empirical patterns in cross-country perspectives (Bueno de Mesquita 2010; Bueno de Mesquita and Shadmehr 2023; Gehlbach, Sonin, and Svolik 2016; Ginkel and Smith 1999; Little 2016). Yet, there are several reasons to think that this phenomenon deserves a closer and deeper study in democratic regimes. First, democracies suit citizens who consider protesting when they hold grievances against political elites or other societal groups, because electoral competition gives elites an incentive to attend to those grievances.

Second, mass protests are costly for incumbent elites in democracies, for two reasons. On the one hand, a massive protest may force politicians to change policies they might not want to alter (Bosi, Giugni, and Uba 2016; Klein and Regan 2018). Gridlocked constitutional provisions that benefit elites are one example, such as the tax code or the electoral system. Massive protests also force the state to spend resources policing them. That usually means maintaining a specialized anti-protest or anti-riot police corps, which is expensive to sustain.

In democracies, the severity of repression is a complex policy written implicitly into the institutional and legal framework. Leaders do not command repressive bureaus to repress citizens in particular ways. Instead, the legal structure sets severity implicitly, by defining how much physical or psychological damage repressive operatives may inflict on dissenting citizens. That structure is slow to change. The model therefore treats the severity policy as chosen before mobilization and held fixed while a protest runs, an assumption I qualify in the next section. Re-

pressive bureaus know the rules of engagement imprinted on the formal institutional framework, which set how much violence they may use against citizens. Politicians weigh costs and benefits carefully when they set severity, because they know that protests are always possible (Nordås and Davenport 2013; Ritter 2014).

In this paper, I develop a game-theoretic model in which the state chooses repression severity and citizens decide whether to protest for policy change. The model has three essential features. First, citizens face a collective action problem when they decide to protest. The protest changes the policy only if participation reaches a critical threshold (Bueno de Mesquita 2010; Bueno de Mesquita and Shadmehr 2023). Second, that critical threshold is not common knowledge. Citizens hold noisy private assessments of how large the protest must be to force the policy change, and they must decide whether to participate under this strategic uncertainty. This is the global games structure. The incomplete information about the fundamental delivers a unique equilibrium in threshold strategies, just as it does in the canonical regime change game, whose complete-information version exhibits multiple equilibria (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). Third, citizens' and state incentives are directly opposed, because the state would prefer to keep the status quo. The state therefore sets a severity of repression that imposes costs on individuals who choose to protest. Higher severity may deter protests, but setting it is costly for the state (Bueno de Mesquita and Shadmehr 2023; Ritter 2014).

The rest of the paper is organized as follows. First, I review the literature that informs the paper's assumptions, including the global games work on protest and regime change that supplies the analytic setup for the citizens' stage. Next, I develop the citizens' participation decision as a global game of policy change, presenting the complete-information benchmark first to make the role of the incomplete information explicit. Then, I integrate the state's choice of severity into a two-stage sequential game. I discuss the model's implications for empirical research. Finally, I present the conclusions of the paper.

2 Literature review.

2.1 State repression, political attitudes, and social movements.

State repression has a long and vast literature. Political repression, or “state” repression, is the use of institutional or physical force, most commonly violent, to subdue or coerce individuals and groups that make claims on or challenge political actors or the status quo (deMeritt 2016). Scholars have documented many empirical regularities and used different theoretical approaches to understand why states repress citizens, and how repression influences political behavior, especially participation in protests. This body of research focuses mostly on authoritarian regimes or flawed democracies. Those regimes enable rulers to deter opposition by violent means, because leaders lack accountability and citizens lack free speech or competitive elections.

Comparative research has persistently found that democratic countries exhibit lower levels of state repression (Davenport 2007a, 2007b; Davenport and Armstrong 2004). Yet state repression is not uncommon in developed democracies, especially violent coercion by state agents, or “repressive operatives.” In fact, state violence, especially police brutality, has become more common in some democracies (Belur 2010; Whitehead, Fair, and Payne 2010; Worden 2015). These expressions of violence are often highly salient and capture public attention. For example, support for the police fell sharply after police killed George Floyd (Reny and Newman 2021). Victims of police repression may also experience anxiety, depression, anger, and fear (Bor et al. 2018; Çelebi, Adam-Troian, and Mahfud 2020; Cooper et al. 2004; J. E. DeVlyder et al. 2017; J. DeVlyder, Fedina, and Link 2020).

Yet not all citizens perceive the use of force by repressive bureaus the same way. In the United States, for example, studies consistently find that conservatives support the police at higher rates than liberals (Ekins 2016; Fingerhut 2017; Hindelang 1974; Zamble & Annesley 1987). Voters may perceive and evaluate state repression differently, to the extent that they have directional motivations and are affective partisans. Attitudes about the police are also racially divided (Ekins 2016; Fingerhut 2017).

One essential question in the literature is how state repression affects social movements. Studies have generally referred to this problem as the “punishment puzzle” (Davenport 2007a). On the one hand, studies have shown that repression discourages mass and peaceful protests but stirs

radicalization (della Porta 1995). In contrast, other studies find that repression incites escalation and radicalization, provoking higher levels of dissent and mobilization (Adam-Troian, Çelebi, and Mahfud 2020; Christopher Michael Sullivan and Davenport 2017).

2.2 Empirical regularities and theoretical foundations.

The literature has identified several determinants that systematically explain state repression, among them regime type, economic development, population size, intrastate civil conflicts, and the degree of grievances and dissent (Davenport 2007b; Nordås and Davenport 2013; Christopher M Sullivan 2016; Christopher Michael Sullivan and Davenport 2017). For example, Nordås and Davenport (2013) argue that the demographic structure of the population is a key determinant of state repression. States respond with more repression where youth bulges are larger, meaning cohorts aged 15 to 24 relative to the overall adult population. States that anticipate more dissent raise repression preemptively, to deter it.

Chenoweth, Perkoski, and Kang (2017) summarize several consistent findings in the literature. First, dissent precedes state repression in most settings, a regularity Davenport (2007a) calls the “law of coercive responsiveness.” Second, democratic countries tend to exhibit lower levels of repression. The authors also discuss the “punishment puzzle” mentioned above. They stress that repression and dissent, whether protest or other tactics, are two complex interlinked processes.

Formal models have approached the theoretical relationship between state repression, protests, and regime or policy change in several ways. No definite consensus has emerged about the best modeling approach. Ritter (2014) notes that models choose different objects of investigation, and that each one, taken alone, misses important strategic features of the relation between repression and dissent. For example, some studies model the determinants of regime change or dissent in general as a process of an ongoing or expected conflict. In one approach, citizens protest to alter the status quo, and their success depends on the state’s concessions. A revolutionary vanguard’s level of violence may signal the level of dissent in society, which provokes citizens to cooperate (Bueno de Mesquita 2010). Other formal models turn to uncertainty and asymmetric information about government types (Pierskalla 2010). A “strong” state may repress citizens at high cost to signal its resolve. Citizens protest over repeated play and learn what type of government they face. Closest to the present paper in timing, De Jaegher and Hoyer (2019) model *preemptive* repression.

A government sets repressive capacity before protest occurs, and dissidents respond by investing in countermeasures. Their model delivers both sides of the punishment puzzle. Repression deters when the benefit of protest is large relative to the cost of countermeasures, and backfires when it is small. It does so, however, in a complete-information coordination game between two pivotal dissidents, where the resulting multiplicity is resolved by the relative size of the equilibria's basins of attraction.

Thus, as Ritter (2014) notes, some models study the conditions that ignite dissent, and others study how the state chooses the severity of repression. But studies miss an important part of the puzzle when they treat these two processes separately. She therefore proposes a model that considers the conditions sparking dissent together with the state's choice of severity. She models the strategic setting as a bargaining problem over policy provision and its effect on incumbent survival. The results show that a higher likelihood of leader survival decreases the level of dissent and repression. Yet the same increase raises the severity of both.

2.3 Global games of protest and regime change.

This paper borrows its analytic setup from the global games literature on coordination under incomplete information (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). The canonical regime change game has a continuum of citizens who simultaneously decide whether to attack a regime. The regime collapses if and only if the mass of attackers exceeds its strength. That strength is an uncertain fundamental, and citizens receive noisy private signals about it. Under complete information about the fundamental, the game exhibits multiple equilibria in an intermediate range of regime strength. The incomplete information selects a unique equilibrium in threshold strategies.

Two recent applications inform the present model directly. Bueno de Mesquita and Shadmehr (2023) develop “a standard global game of regime change.” A continuum of citizens receive noisy private signals $x_i = \theta + \sigma \varepsilon_i$ about the regime's strength θ , and the rebellion succeeds if and only if the fraction of rebels exceeds θ . Their complete-information benchmark (their Proposition 1) shows that for intermediate strength “both equilibria coexist.” They note that this multiplicity “makes empirical prediction difficult” and renders outcomes “insensitive to parameters of the model like the costs of rebellion.” Their incomplete-information analysis (their Proposition 2)

delivers a unique equilibrium, in which the rebellion succeeds if and only if the fundamental is below a threshold. That threshold equals $1 - c$ in their psychological rewards setting, where c is the cost of participation. They adopt “the standard conception of state repression as any action by the state that ‘raises the contenders’ cost of collective action’ (Tilly 1978, 100),” and they represent more repression as a higher c . They add a first stage in which the government chooses its repression level by minimizing the sum of the regime change threshold and a convex cost of repressive capacity. The present model adopts one ingredient of their psychological rewards setting. The participation reward depends on the movement’s success. Participants enjoy the psychological reward of having taken part only if the rebellion succeeds, an idea that builds on Wood’s (2003) notion of pleasure-in-agency. This success-contingency generates the strategic complementarity that the global game resolves.

Little (2016) distinguishes two coordination problems that potential protesters face. The first is “political coordination,” deciding whether to participate without knowing how many others are dissatisfied enough to join. The second is “tactical coordination,” deciding when, where, and how to protest. His political coordination model also has a continuum of citizens with heterogeneous assessments and equilibria in cutoff strategies. He shows that better tactical information “almost always lowers the effective cost of protest, leading to larger (and more cohesive) antiregime action.” This distinction matters for the present model, because communication technology conveys information about the critical threshold that citizens face here. I return to the point in the empirical implications.

3 Modeling protests, repression, and policy change.

In this section, I develop a model of citizens who decide to protest to change a disliked policy. The citizens face a coordination and collective action problem. The protest changes the policy only if citizens’ participation reaches a given threshold. The threshold is an uncertain fundamental, and citizens hold noisy private assessments of it. On the other side, the state would prefer not to change the policy. To influence participation, it can set a severity of repression that inflicts costs on those who protest. Both are costly for the state, however. Repression consumes resources, since producing a given level of it means producing a given number of victims, and setting severity

invites backlash from public opinion. I begin with the citizens' participation problem and its complete-information benchmark. Next, I solve the global game. Then, I propose the two-stage sequential game in which the state sets severity first, and I inspect the strategic implications of the model.

3.1 The citizens' participation problem.

There is a continuum of citizens of mass one, indexed by $i \in [0, 1]$. Each citizen chooses between participating in the protest (P) or not (NP). Let $n_i \in \{0, 1\}$ indicate participation, and let $N^* \in [0, 1]$ denote the realized mass of participants. Protesters want to change a policy so gridlocked that ordinary representative channels struggle to move it. The protest is successful if participation reaches a critical threshold τ , so the policy changes if and only if $N^* \geq \tau$.

The payoff primitives are as follows. A change in the policy provides a collective and indivisible benefit B_p to all individuals in the society (i.e., a public good). Individuals who participate may get an expressive or emotional benefit d_p . They may feel, for example, that fighting for a just cause is good in and of itself. Participants also pay the expected cost of repression $\gamma_r \cdot c_r$, where c_r is the severity of repression (the disutility inflicted on a repressed protester) and $\gamma_r \in [0, 1]$ is the probability that a participant is a victim of state repression. Both c_r and γ_r are policy variables for the state.

Two features of the continuum setting must be made explicit, because they discipline how the primitives enter the participation margin.

First, no individual citizen is pivotal. With finitely many players, Δp_i would carry the public good benefit B_p into the individual calculus. That term is the change in the probability of reaching the threshold induced by one's own participation. In the continuum, a single citizen has measure zero, so $\Delta p_i = 0$. The citizen receives B_p when the protest succeeds whether she participated or not, and B_p cancels from the comparison between participating and abstaining. The free-riding logic is total here. This is not a defect of the analytic setup but a substantive statement that the setup forces on us. In a mass protest, the policy prize itself cannot move the marginal participant, because her participation does not detectably move the prize.

Second, the expressive benefit must be success-contingent for participation to respond to the prospects of success at all. I follow the psychological rewards setting of Bueno de Mesquita and

Shadmehr (2023), which builds on Wood’s (2003) pleasure-in-agency. A participant enjoys the expressive reward d_p only if the movement succeeds. That satisfaction arrives only when the change actually happens. This success-contingency is a substantive assumption, and it generates the coordination problem. Were d_p unconditional, participation would reduce to the comparison of d_p against $\gamma_r c_r$, independent of any belief about others, and the strategic structure would disappear.

With these two features, the net payoff of participating relative to abstaining, for citizen i , is

$$u_i(P) - u_i(NP) = \mathbb{1}\{N^* \geq \tau\} \cdot d_p - \gamma_r \cdot c_r,$$

since $B_p \cdot \mathbb{1}\{N^* \geq \tau\}$ enters both sides and cancels. A citizen therefore participates when her belief that the protest will succeed, times the expressive reward, exceeds the expected repression cost.

3.2 The complete-information benchmark.

Suppose first, as a benchmark, that τ is common knowledge. Assume $0 < \tau \leq 1$, the interesting range, and $\gamma_r c_r < d_p$, so that participation is worthwhile when success is assured. Then two pure strategy equilibria coexist, and the incomplete-information version removes this multiplicity (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). If every citizen participates, $N^* = 1 \geq \tau$, the protest succeeds, and each participant receives $d_p - \gamma_r c_r > 0$. No citizen gains by deviating, since abstaining forfeits $d_p - \gamma_r c_r$ while leaving success (and hence B_p) unchanged. If no citizen participates, $N^* = 0 < \tau$, the protest fails, and a lone deviator would pay $\gamma_r c_r$ for a failed protest. Universal abstention is therefore also an equilibrium. If instead $\gamma_r c_r > d_p$, universal abstention is the unique equilibrium, because participation is dominated even under assured success.

This benchmark mirrors the complete-information benchmark of Bueno de Mesquita and Shadmehr (2023, their Proposition 1), and it inherits the two problems they identify. The multiplicity makes prediction difficult, and the equilibrium set is insensitive to the parameters, including the repression policy itself. One could respond by selecting an equilibrium, for instance the mixed strategy equilibrium, and reading its participation probabilities as frequencies. I instead take seriously that the critical threshold is a variable citizens cannot observe. The size a protest must reach to force a policy change is written nowhere, unlike the threshold to win an election

under plurality rule, which is common knowledge. How citizens form beliefs about τ is therefore central to any behavioral prediction.

3.3 The global game.

Let the critical threshold τ be drawn by nature and unobserved by citizens. Citizens share an improper uniform prior over $\mathbb{R}$, the standard device in this literature to isolate strategic uncertainty (Morris and Shin 2003; Bueno de Mesquita and Shadmehr 2023).¹ Each citizen receives a noisy private signal

$$x_i = \tau + \sigma \varepsilon_i,$$

where the ε_i are i.i.d. with a log-concave density f and c.d.f. F with full support on $\mathbb{R}$, independent of τ , and $\sigma > 0$ scales the noise. A citizen with a low signal believes the required crowd is small and success likely. A citizen with a high signal believes success is out of reach.

A strategy is a mapping from signals to participation. The participation payoff is decreasing in the (believed) threshold, so cutoff strategies are the natural place to look for equilibria. A citizen participates if and only if $x_i < \hat{x}$. Given such a cutoff, the mass of participants at true threshold τ is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ . The protest therefore succeeds if and only if τ is below a unique critical value $\hat{\tau}$. The equilibrium is characterized by the same two conditions as in the canonical regime change game (Bueno de Mesquita and Shadmehr 2023, their equation (1)):

$$\text{(belief consistency / critical mass):} \quad F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = \hat{\tau},$$

$$\text{(indifference of the marginal citizen):} \quad \Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) \cdot d_p = \gamma_r c_r.$$

The first condition says that at the critical threshold the crowd is exactly critical. The second says that the citizen holding exactly the cutoff signal is indifferent between participating and

¹The improper prior delivers exact closed forms. The same characterization obtains for any smooth proper prior in the limit as the signal noise becomes small.

abstaining. Under the improper uniform prior, the posterior of τ given $x_i = \hat{x}$ is $\hat{x} - \sigma\epsilon$, so

$$\Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) = 1 - F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = 1 - \hat{\tau},$$

where the last equality substitutes the belief consistency condition. The indifference condition then collapses to $(1 - \hat{\tau})d_p = \gamma_r c_r$, and the equilibrium success threshold is

$$\hat{\tau}(c_r) = 1 - \frac{\gamma_r c_r}{d_p}, \quad \text{for } \frac{\gamma_r c_r}{d_p} \in [0, 1]. \quad (1)$$

This is the exact analog of the threshold $\theta^p = 1 - c$ in Bueno de Mesquita and Shadmehr (2023, their equation (2) and Proposition 2), with the normalized participation cost $\gamma_r c_r / d_p$ in the role of c . Proposition 1 collects the characterization.

Proposition 1 (Equilibrium of the citizens' stage). *Fix a severity policy with $\gamma_r c_r / d_p \in (0, 1)$. The citizens' stage has a unique equilibrium in monotone strategies. Citizen i participates if and only if $x_i < \hat{x} = \hat{\tau} + \sigma F^{-1}(\hat{\tau})$, and the protest succeeds if and only if $\tau \leq \hat{\tau} = 1 - \gamma_r c_r / d_p$. If $\gamma_r c_r \geq d_p$, no citizen participates and the protest fails against every $\tau > 0$; if $c_r = 0$, every citizen participates and $\hat{\tau} = 1$. Standard arguments extend uniqueness beyond the monotone class.²*

The full proof is in the appendix. Three remarks.

First, the multiplicity is gone, and the model makes one prediction for every value of the repression policy. The protest succeeds if and only if the realized threshold satisfies $\tau \leq \hat{\tau}(c_r)$. The comparative statics of the model are therefore unconditional statements, not statements contingent on which equilibrium is played or on a selected mixed strategy.

Second, the repression policy now moves the outcome smoothly. The deterrence derivative is

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{\gamma_r}{d_p}.$$

Each unit of severity shrinks the set of fundamentals at which the protest succeeds at a constant rate, inversely proportional to the expressive reward. Repression is blunt against a strongly motivated citizenry (high d_p), and sharp against a weakly motivated one. This is the global games

²See Morris and Shin (2003) and the technical discussion in Bueno de Mesquita and Shadmehr (2023, their footnote 8).

rendering of a familiar intuition, that intrinsic motivation erodes the effectiveness of severity.

Third, the deterrence margin is governed by d_p alone, not by $d_p + B_p$. Full deterrence would require severity to climb over the policy prize itself in a small-numbers setting where participation is pivotal, since the decisive citizen weighs B_p directly. De Jaegher and Hoyer (2019) study such a setting, where the protest occurs only if both dissidents participate. Here, full deterrence ($\hat{\tau} = 0$) is reached already at $c_r = d_p/\gamma_r$. Severity never needs to climb over B_p , because the free-riding of the continuum has already removed B_p from the participation margin. This is a substantive, testable implication of the mass-mobilization margin, discussed further below.

For the state's problem it is convenient to work in the small-noise limit $\sigma \rightarrow 0$. In that limit the cutoff $\hat{x}$ converges to $\hat{\tau}$ and participation becomes sharp: $N^*(\tau) \rightarrow 1$ for $\tau < \hat{\tau}$ and $N^*(\tau) \rightarrow 0$ for $\tau > \hat{\tau}$. Protests that succeed are massive; protests that fail do not materialize. This stark prediction is an artifact of vanishing noise, and with $\sigma > 0$ participation falls continuously in τ . The limit is used only to keep the state's objective in closed form.

3.4 The state's problem.

The state moves first and sets the repression severity policy, anticipating the unique equilibrium of the citizens' stage.

Assumption 1 (Severity is fixed within a protest episode). The state chooses c_r before mobilization begins and does not revise it while the protest unfolds. Severity is chosen freely between episodes; it is sticky within one.

Rules of engagement are written into legal structure and standing procedure, as described above, and changing them takes longer than a protest lasts. The assumption also makes the game sequential. Without it the state and the citizens move simultaneously, and severity loses its commitment value.

De Jaegher and Hoyer (2019) take the other branch. Their government observes how far dissidents have invested in countermeasures and adapts its repression level accordingly, and the delay before it can do so is a parameter of their analysis. This model is the limit of that setup in which the delay exceeds the length of a protest episode. Which limit applies is an empirical question about the institution. Severity written into legal and doctrinal frameworks moves slowly.

Repressive capacity that can be surged within days does not, and there the fixed-policy assumption would be the wrong one. I fix $\gamma_r = 1$ to focus on c_r as the policy variable, and I assume the following cost function for the production of repression:

$$f(c_r, N^*) = \theta_0 \cdot N^* + \theta_1 c_r^2,$$

where θ_0 is the marginal cost of repressing one additional protester (e.g., costs in repressive inputs or equipment) and θ_1 scales the convex public opinion cost of severity, whose marginal cost is $2\theta_1 c_r$. The state keeps B_s if the protest fails and loses it if the protest succeeds. Throughout I assume $B_s > 0$, $\theta_0 \geq 0$, and $\theta_1 > 0$.

The state evaluates its policy *ex ante*, before nature draws τ . For this evaluation the improper prior is not usable, and I endow the state with a proper planning prior $\tau \sim U[0, 1]$, following the same logic by which Bueno de Mesquita and Shadmehr (2023) give their government a proper distribution over the fundamental at its investment stage.³ Under this prior and the small-noise limit, the probability that the protest succeeds is $\Pr(\tau \leq \hat{\tau}) = \hat{\tau}(c_r)$. The expected mass of protesters is $E[N^*] = \hat{\tau}(c_r) \cdot 1 = \hat{\tau}(c_r)$, since in the limit the protest is total exactly when it succeeds. The state's expected utility is

$$u_S = (1 - \hat{\tau}(c_r))B_s - \theta_0 \hat{\tau}(c_r) - \theta_1 c_r^2 = \frac{c_r}{d_p} B_s - \theta_0 \left(1 - \frac{c_r}{d_p}\right) - \theta_1 c_r^2,$$

substituting equation (1) with $\gamma_r = 1$. The objective is strictly concave in c_r , so the first order condition characterizes the optimum on the relevant domain $c_r \in [0, d_p]$. Beyond d_p , additional severity produces no further deterrence, since $\hat{\tau}$ is already zero, and only adds public opinion costs. The first order condition is

$$\frac{B_s + \theta_0}{d_p} = 2\theta_1 c_r,$$

which says that the marginal benefit of severity equals the marginal public opinion cost. That benefit is deterrence at rate $1/d_p$, valued at the status quo prize, plus the production cost saved

³Their government minimizes $\theta^j(c) + C(c)$; the uniform planning prior makes the probability of regime change equal to the threshold itself, which is the same simplification used here.

per deterred protest. Solving,

$$c_r^* = \min \left\{ \frac{B_s + \theta_0}{2\theta_1 d_p}, d_p \right\}. \quad (2)$$

Proposition 2 (The optimal severity policy). *The state's objective is strictly concave on $[0, d_p]$ and severity above d_p is never optimal, so the optimum is unique and given by (2). The solution is interior when $B_s + \theta_0 < 2\theta_1 d_p^2$, and otherwise the state sits at the full-deterrence corner $c_r^* = d_p$, at which the protest never succeeds. The two coincide at equality. In either case $c_r^* > 0$.*

The full proof is in the appendix, and Panel A of Figure 1 maps the two regimes in the parameter space. The equilibrium success threshold is also the probability of policy change and the expected protest size. At the interior optimum it is

$$\hat{\tau}^* = 1 - \frac{B_s + \theta_0}{2\theta_1 d_p^2}.$$

3.5 Comparative statics.

The comparative statics at the interior solution are, without exception, unambiguous.

Proposition 3 (Comparative statics). *At the interior solution,*

$$\begin{aligned} \frac{\partial c_r^*}{\partial B_s} &= \frac{1}{2\theta_1 d_p} > 0, & \frac{\partial c_r^*}{\partial \theta_0} &= \frac{1}{2\theta_1 d_p} > 0, & \frac{\partial c_r^*}{\partial \theta_1} &= -\frac{B_s + \theta_0}{2\theta_1^2 d_p} < 0, \\ \frac{\partial c_r^*}{\partial d_p} &= -\frac{B_s + \theta_0}{2\theta_1 d_p^2} < 0, & \frac{\partial c_r^*}{\partial B_p} &= 0, \end{aligned}$$

and the equilibrium success probability $\hat{\tau}^$ is decreasing in B_s and θ_0 , increasing in θ_1 and d_p , and independent of B_p .*

Figure 1 illustrates these comparative statics by region of the parameter space, plotting the latent interior solution against the effective policy in each regime.

The signs admit direct interpretation. An increase in the production cost parameter θ_0 raises the chosen severity. If repressing each protester is dearer, the state deters harder to face fewer of them. An increase in the public opinion cost parameter θ_1 lowers severity. An increase in the intrinsic motivation d_p lowers severity, because motivated citizens are expensive to deter. The state's retreat does not fully compensate, however, since $\hat{\tau}^*$ is increasing in d_p , so at an interior

Region-conditional comparative statics, global-games version
 $(\theta_1 = 1, d_p = 3 \text{ except panel C; } B_s = 3 \text{ in panel C; dotted = latent interior FOC, solid = effective policy})$

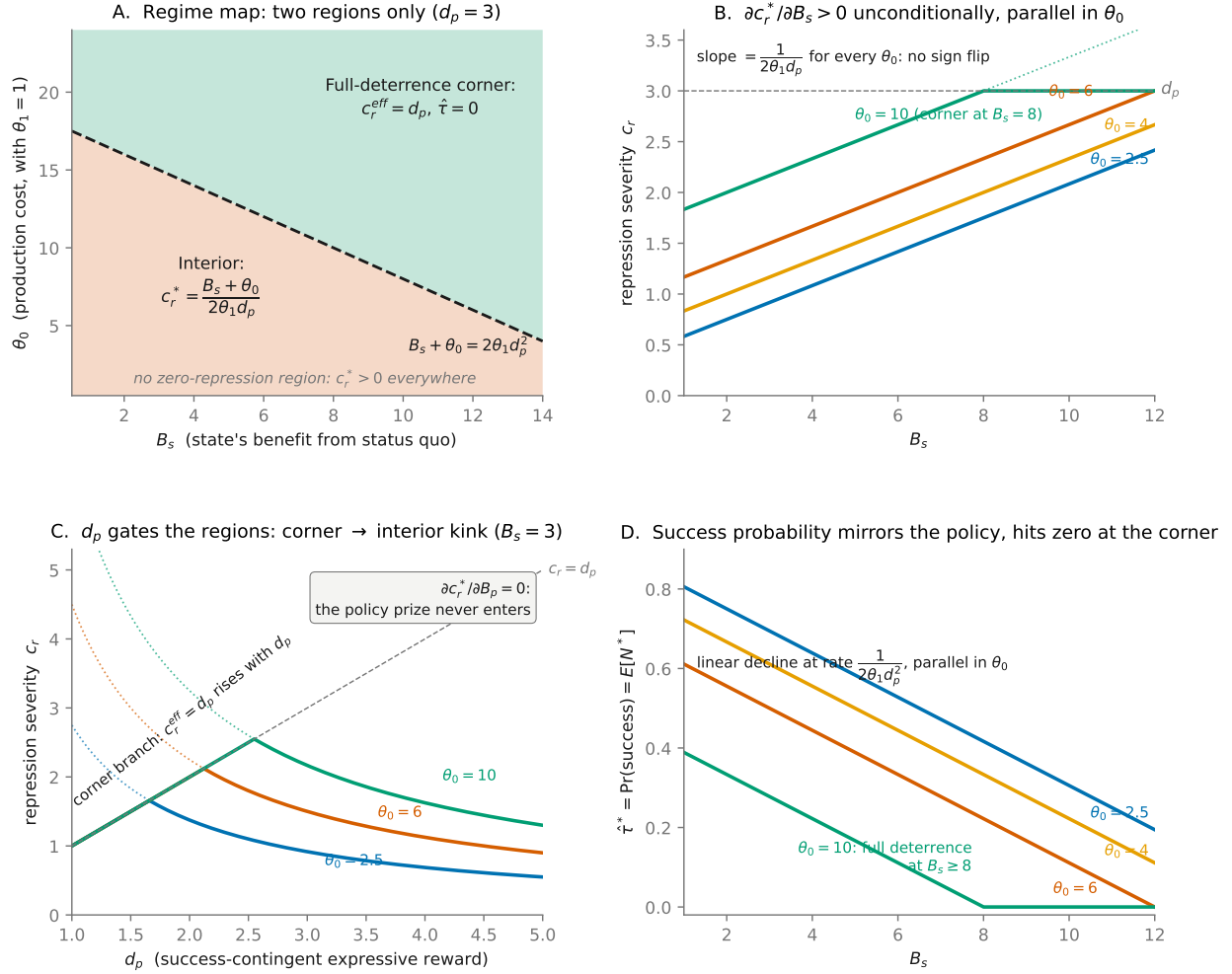


Figure 1: Region-conditional comparative statics of the repression severity policy ($\theta_1 = 1$; $d_p = 3$ except in panel C; $B_s = 3$ in panel C). Panel A maps the two regions of the parameter space, the interior solution and the full-deterrence corner, separated by $B_s + \theta_0 = 2\theta_1 d_p^2$. There is no zero-repression region. Panel B shows that c_r^* rises linearly in B_s with slope $1/(2\theta_1 d_p^2)$ for every θ_0 , flattening at the corner $c_r = d_p$. Panel C shows the kinked, non-monotone effect of d_p : the effective policy rides the corner $c_r = d_p$ upward while full deterrence is cheap, then switches to the declining interior solution. Panel D plots the equilibrium success probability $\hat{\tau}^*$, which declines linearly in B_s at rate $1/(2\theta_1 d_p^2)$ and reaches zero at the corner. In panels B and C, dotted curves are the latent interior first order condition and solid curves are the effective policy.

optimum more motivated citizenries both face milder repression and succeed more often. The comparison reverses at the corner, where severity tracks d_p one for one and the protest never succeeds.

Two features of these results deserve emphasis.

First, the status quo prize B_s has an unambiguously positive effect, with no parametric condition attached. In coordination models resolved by equilibrium selection, comparative statics of this kind are typically regime-conditional, holding in some regions of the parameter space and reversing in others. In De Jaegher and Hoyer (2019), for instance, repression deters when $V > 2c$ and backfires when $V < 2c$. Here the unique threshold equilibrium makes the deterrence technology the linear relation (1), and the marginal value of severity is monotone in the stakes. Panels B and D of Figure 1 display this monotone response and its mirror image in the success probability.

Second, B_p never enters the state's problem. The severity policy does not reckon with the policy prize at all. Positive severity is optimal whenever $B_s + \theta_0 > 0$, that is, always, and the size of the prize the citizens would win is irrelevant to how hard the state represses. Panel C of Figure 1 shows the parameter that gates the policy instead. The effective severity is non-monotone in d_p , riding the full-deterrence corner upward before switching to the declining interior solution. This is the model working as intended, and it delimits its scope. The model instantiates a mass setting in which individual participation is not pivotal. In a small-numbers setting, the pivotal calculus of a small group carries B_p into the participation margin and, with it, into the severity policy. Which margin governs actual protest participation is an empirical question, and the sharp difference in the role of B_p across the two margins converts that question into a testable one.

3.6 An extension: Repression, protests, and emotions.

I now adapt the model to introduce the role of emotions in the game's dynamics. Let the expressive reward be a function of the emotional state “ e ” of an individual, $d_p(e) = d_0 + e(c_r)$. In democracies, open media lets citizens learn how repressive the average state operative is. Citizens may get angry when they learn that severity is high, so that $e'(c_r) > 0$ for low levels of c_r . Yet for high or extreme levels of c_r , anticipatory fear may dominate and $e'(c_r) < 0$.

In the global games framework this channel acquires a precise transmission mechanism: the

equilibrium threshold becomes

$$\hat{t}(c_r) = 1 - \frac{\gamma_r c_r}{d_0 + e(c_r)},$$

so the emotional response enters the citizens' stage through the denominator of the normalized participation cost. I develop the extension in three steps. First, I derive the general condition under which severity deters or backfires. Next, I solve the state's problem when anger responds linearly to severity. Then, I consider an outrage specification in which the anger response accelerates, which produces a backfire region and caps the state's optimal severity.

Let us fix $\gamma_r = 1$ and differentiate the threshold with respect to the severity policy. We get

$$\frac{\partial \hat{t}}{\partial c_r} = - \frac{d_0 + e(c_r) - c_r e'(c_r)}{(d_0 + e(c_r))^2}.$$

Define the elasticity of motivation with respect to severity as $\varepsilon(c_r) = c_r e'(c_r)/d_p(c_r)$.

Proposition 4 (Deterrence and backlash). *Severity deters, $\partial \hat{t}/\partial c_r < 0$, if and only if $\varepsilon(c_r) < 1$; it backfires, $\partial \hat{t}/\partial c_r > 0$, if and only if $\varepsilon(c_r) > 1$.*

In other words, whether repression discourages or recruits is not a psychological primitive but an elasticity condition on the anger response. This gives the “punishment puzzle” (Davenport 2007a) a precise formulation. The deterrence and backlash findings that coexist in the empirical literature (della Porta 1995; Christopher Michael Sullivan and Davenport 2017; Adam-Troian, Çelebi, and Mahfud 2020) correspond to the two sides of the same inequality.

Let us begin with the linear case, $e(c_r) = \alpha c_r$ with $\alpha \geq 0$, so that $d_p(c_r) = d_0 + \alpha c_r$. The elasticity satisfies $\varepsilon(c_r) = \alpha c_r/(d_0 + \alpha c_r) < 1$ for every severity level, so linear anger never reverses deterrence; it dampens it, given that $\partial \hat{t}/\partial c_r = -d_0/(d_0 + \alpha c_r)^2$. The threshold is also bounded below. As c_r grows, $\hat{t}$ approaches $1 - 1/\alpha$. Thus, if $\alpha \geq 1$, full deterrence is unattainable at any finite severity, and an angry-enough citizenry retains a strictly positive probability of policy change no matter how harsh the policy. If $\alpha < 1$, full deterrence requires $c_r = d_0/(1 - \alpha)$, which grows without bound as α approaches one.

The state's problem remains strictly concave, and the first order condition becomes $(B_s + \theta_0)d_0 = 2\theta_1 c_r(d_0 + \alpha c_r)^2$, which has a unique positive solution.⁴ The full-deterrence corner is

⁴The left-hand side is a positive constant and the right-hand side is strictly increasing from zero, so the two cross exactly once. The derivations for this section are collected in the appendix.

optimal if and only if $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$, which collapses to the condition of the baseline model at $\alpha = 0$ and becomes impossible to satisfy as α approaches one. Two comparative statics follow. First, $\partial c_r^*/\partial \alpha < 0$, so the state tempers the severity of repression when facing a more anger-reactive citizenry. Second, $d\hat{\tau}^*/d\alpha > 0$, so the equilibrium probability of policy change increases in anger-reactivity through both channels, directly, because anger blunts deterrence, and strategically, because the state represses less.

Proposition 5 (Linear anger). *Let $e(c_r) = \alpha c_r$ with $\alpha \geq 0$. Then: (i) $\varepsilon(c_r) < 1$ at every severity and $\partial \hat{\tau}/\partial c_r = -d_0/(d_0 + \alpha c_r)^2 < 0$; (ii) $\hat{\tau}(c_r) > 1 - 1/\alpha$ at every finite severity, so full deterrence is unattainable whenever $\alpha \geq 1$, while for $\alpha < 1$ it requires $c_r = d_0/(1 - \alpha)$; (iii) the state's objective is strictly concave, the optimum is unique, and the full-deterrence corner is optimal if and only if $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$; (iv) at the interior optimum, $\partial c_r^*/\partial \alpha < 0$ and $d\hat{\tau}^*/d\alpha > 0$.*

Next, let us consider an outrage specification, $e(c_r) = \alpha c_r^k$ with $k > 1$, in which the anger response accelerates with severity. The idea is that extreme severity is qualitatively different from routine crowd control. A police killing that saturates media coverage recruits participants in a way that tear gas does not (Reny and Newman 2021). Now $\varepsilon(c_r)$ is increasing in c_r and crosses one at $\tilde{c} = (d_0/(\alpha(k - 1)))^{1/k}$, so the threshold $\hat{\tau}$ is U-shaped. Severity deters below $\tilde{c}$ and backfires above it. Two consequences follow. First, the backfire point caps the state's policy, and the optimal severity never exceeds $\tilde{c}$, given that pushing beyond it raises both the probability of policy change and the costs of repression. Second, if $\hat{\tau}(\tilde{c}) = 1 - \tilde{c}(k - 1)/(d_0 k) > 0$, full deterrence is unattainable at any severity, and the best the state can do is push the probability of policy change down to its floor at the backfire point.

Proposition 6 (Outrage). *Let $e(c_r) = \alpha c_r^k$ with $\alpha > 0$ and $k > 1$. Then: (i) $\hat{\tau}$ is U-shaped in c_r , with its minimum at $\tilde{c} = (d_0/(\alpha(k - 1)))^{1/k}$, where it takes the value $1 - \tilde{c}(k - 1)/(d_0 k)$; (ii) the optimal severity satisfies $c_r^* \leq \tilde{c}$; (iii) if $\hat{\tau}(\tilde{c}) > 0$, no severity attains full deterrence. When $\hat{\tau}(\tilde{c}) < 0$, the threshold clamps at zero over an interval around $\tilde{c}$.*

Finally, fear is the mirror case and requires little development. If $e'(c_r) < 0$, then $\varepsilon(c_r) < 0$, deterrence is amplified, and the qualitative structure of the baseline model is preserved with a steeper deterrence derivative. The interesting asymmetry is that anger changes the structure of

the state's problem, bounding its policy and possibly its payoff, while fear only rescales it. A full nonmonotone specification, in which anger dominates at low severity and fear at extreme severity, combines the two regimes. Its analysis adds no closed forms, so I do not pursue it here.

4 Empirical implications.

The game-theoretic model developed above offers several empirical implications. I suggest some of them and propose ways to estimate the empirical relation between protests and repression.

First, the model suggests that a naive regression between protests and repression will give wrong estimates, whether one estimates the effect of protests on repression or the reverse. This holds even with many meaningful controls for omitted variables. This happens because repression and the size of protests are simultaneously defined. Hence, empirical research should consider using structural estimation models or models that exploit some natural experiment, such as instrumental variables. The incomplete-information structure strengthens the case for the structural route. Because the equilibrium is unique, the model delivers point predictions, a likelihood can be written down for observed outcomes, and the parameters $(d_p, \theta_0, \theta_1, B_s)$ are identified from variation in repression severity and protest success without an auxiliary equilibrium selection assumption. Multiplicity is what makes structural estimation of coordination models treacherous, and the analytic setup removes it.

Second, the production of repression is costly in democracies, a point the model makes explicit. Overall repression has production costs even if we are unsure about the magnitude of public opinion costs, among them spending on police trucks, ammunition, armored water cannon, and police buildings. The severity of repression is therefore a strategic choice, because fiscal spending has opportunity costs and incumbents could have used those resources to favor voters directly.

An increase in public opinion marginal costs may reduce the severity of repression, which is not a surprising result. Such an increase, which is common in democracies, could therefore induce larger protests or more participation in general.

Suppose empirical researchers can track exogenous shocks unrelated to the determinants of protest participation, which influence either the production costs of repression (in the model, θ_0)

or the public opinion costs (in the model, θ_1). They could then build a measure of these shocks and exploit it in an instrumental variables model.

The model's distinctive predictions supply an additional, sharper test. The severity of repression should respond positively to the incumbent's stakes in the status quo (B_s) everywhere in the parameter space. It should not respond to the size of the prize citizens would obtain from the policy change (B_p). The second prediction is the signature of the mass-mobilization margin. It holds because no participant is pivotal, and a pivotal calculus would restore B_p to the participation margin.

Existing cross-national evidence is consistent with the first of these. Klein and Regan (2018) define the concession cost of a protest as the political cost of altering the status quo policy in response to protesters' demands, which is the empirical counterpart of B_s . Across 161 countries between 1990 and 2014, they find that governments facing high concession costs respond coercively. Their dependent variable is the mode of the response, coercion or accommodation, and not the severity of repression on a continuous scale, so the correspondence is conceptual. The sign, however, is the one the model predicts everywhere in the parameter space. This pattern should hold where protests are massive and individual pivotality is negligible, but not where groups are small, organized, and pivotal. A design would discriminate between the two margins by comparing repression severity across policy disputes that differ in the dispersion of the prospective benefits, holding the incumbent's stakes fixed.

The informational structure also generates implications of its own. Citizens' behavior depends on their beliefs about the critical threshold, and those beliefs respond to public information about expected turnout, past protest sizes, and logistical focal points. This connects directly to Little's (2016) distinction between political and tactical coordination. Information operates on the very fundamental τ of the present model when it improves citizens' assessments of the crowd required and of the crowd forthcoming. He finds that better tactical information systematically increases protest, which suggests that communication technology should compress the noise σ and sharpen the participation cascade around the threshold. Exogenous variation in connectivity or in media coverage of expected turnout could test whether protest outcomes become more bimodal, either massive or absent, as the model predicts they should when strategic uncertainty shrinks.

However, one obvious and nontrivial challenge is how to measure any of these variables. First,

protest size seems the least difficult to measure, though it is not easy. Several data sets code protest event characteristics by analyzing media content. Some record the size of participation directly.⁵ Survey data can also approximate the intrinsic motivations to participate in protests (d_p) at the individual level, through question items that gauge political involvement or attitudes towards political participation. Under the success-contingent interpretation adopted here, the relevant survey construct is not generic expressiveness but the anticipated satisfaction of having contributed to a victory, which is closer to efficacy and pleasure-in-agency item batteries. The variable B_p exits the participation margin here, which is a modest empirical convenience of this modeling strategy. It is an unobserved construct and among the hardest in the model to measure.

We may be able to proxy for the state's marginal costs of repression. For example, one could measure the marginal costs of producing repression (θ_0) by the average wage of repressive operatives, or by the percent of GDP the state devotes to repressive bureaus. We might also use the distance from protest events to police stations, or to the relevant repressive bureau offices, as an instrument for the level of repression in a given protest episode. The idea is that a greater distance raises the cost of mobilizing repressive operatives. Hence there is an incentive to increase the severity of repression. We would then predict a smaller protest where severity is set high.

The conditions of the competitive political environment may proxy for the marginal costs related to public opinion. This variable also requires attention to the institutional framework at the national or subnational level. For example, incumbents at the national level could face different effects where repressive operatives are managed subnationally or locally. The idea is that repression may be more costly for the incumbent where the electoral setting is competitive.

Next, I briefly discuss ways in which we could measure the repression severity policy in democracies. In this case, data is indeed scarce. Efforts are growing, such as the Mapping Police Violence project in the United States, but coverage remains thin. We can still ask how to approach this construct with better data. I see two ways to measure it. First, we could calculate the rate of victims of repression over the population as some studies have done. The open question in this approach would be how to choose the appropriate geographical area to define the base of the rate. Second, if we had such data, we could elaborate on a measure of physical or psychological trauma for the

⁵For example, the Mass Mobilization Protest Data project, the Armed Conflict Location & Event Data Project, the Social Conflict Analysis Database, etc.

average victim of repression.

Finally, survey data or experiments could measure the relation between the severity of repression and the emotional state it elicits in citizens who may or may not plan to participate in future protests. Such an assessment should test whether different levels of severity have nonlinear effects on individuals' emotional states. For example, one may probe subjects with different and explicit consequences of repression victimization under a survey experiment. Then, one would measure their emotional state. Finally, one would ask individuals' dispositions or attitudes to participate in protests. In the extension, the object to estimate is the elasticity of intrinsic motivation with respect to severity. The model predicts deterrence where it is below one and backlash where it is above one, so the design just described doubles as a direct measurement of the backfire point.

5 Conclusion.

Using a game-theoretic model, this paper explored the relationship between repression and protests motivated by policy change in democratic regimes. I build a simple sequential game that considers citizens' decisions to protest and the state's decision to set a severity of repression that deters participation. The citizens' stage is a global game of policy change. The critical threshold that protests must reach is an uncertain fundamental, and citizens hold noisy private assessments of it. The resulting incomplete information selects a unique equilibrium in threshold strategies, where the complete-information benchmark exhibits multiple equilibria.

In this setting, the state moves first and defines a degree of severity of repression policy, which sets the individual costs for the repressed protester. The state would prefer that protests be minimal to keep the policy status quo unchanged, yet producing repression is costly for the state. In the second stage, citizens decide whether to protest under strategic uncertainty about the critical threshold. The protest succeeds, and the incumbent politicians are forced to change the policy, exactly when the realized threshold falls below an equilibrium cutoff that the state's severity policy shifts down at a rate inversely proportional to citizens' intrinsic motivation.

The model offers several insights. First, a unique equilibrium interlinks the severity of repression with the prospects of protest, so the model's comparative statics are unconditional. Severity rises with the incumbent's stakes in the status quo and with the marginal production cost of re-

pression, and falls with the public opinion cost of severity and with citizens' intrinsic motivation. Second, the policy prize drops out of the participation margin altogether. In a mass protest no citizen is pivotal, so the prize cannot move the marginal participant. This prediction contrasts with the pivotal logic of small-numbers settings, and that contrast is a testable difference between mass and small-numbers mobilization rather than a defect of the model. Third, severity rises with the marginal cost of policing an additional protester, because deterrence economizes on the production of repression. Public opinion bounds that response, since severity falls as the marginal cost of backlash rises.

Several open questions remain and suggest extensions to the model. First, the emotions extension develops the anger channel for linear and accelerating responses, showing that backlash is an elasticity condition and that outrage caps the state's optimal severity. A full nonmonotone specification, in which fear overtakes anger at extreme severity, remains open. Second, a repeated version would connect the model to learning dynamics over protest waves, with citizens observing protest outcomes and updating about a persistent threshold. Third, the state itself could be given instruments over the information structure, censoring or publicizing turnout information, which the global games literature on communication technology suggests is a first-order policy margin (Little 2016). The biggest empirical challenge is how to measure the model's variables. Production costs could be proxied by the average wage of repressive operatives or by the share of GDP devoted to repressive bureaus. Severity could be measured by the physical or psychological trauma of the average repressed protester. A more feasible option is simply the number of repressed protesters as a rate of the local population. Researchers should continue to study repression and protest, which have become salient and influential in current politics, even in developed democracies.

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6 Appendix.

6.1 A preliminary lemma.

Lemma 1 (Posterior beliefs under the improper prior). *Suppose τ has the improper uniform prior on $\mathbb{R}$ and $x_i = \tau + \sigma \varepsilon_i$ with $\varepsilon_i \sim f$. Then the posterior density of τ given x_i is $f((x_i - \tau)/\sigma)/\sigma$, and for every evaluation point t ,*

$$\Pr(\tau \leq t \mid x_i) = 1 - F\left(\frac{x_i - t}{\sigma}\right).$$

Proof. By Bayes’ rule with a flat prior, the posterior density of τ given x_i is proportional to the likelihood of x_i given τ , which is $f((x_i - \tau)/\sigma)/\sigma$. This function integrates to one over $\tau \in \mathbb{R}$: substituting $u = (x_i - \tau)/\sigma$, so that $d\tau = -\sigma du$,

$$\int_{-\infty}^{\infty} \frac{1}{\sigma} f\left(\frac{x_i - \tau}{\sigma}\right) d\tau = \int_{-\infty}^{\infty} f(u) du = 1,$$

so the proportionality constant is one and the likelihood is itself the posterior density. For the distribution function, integrate over $\tau \leq t$ with the same substitution; the bound $\tau = -\infty$ maps to

$u = \infty$ and $\tau = t$ maps to $u = (x_i - t)/\sigma$, so

$$\Pr(\tau \leq t \mid x_i) = \int_{-\infty}^t \frac{1}{\sigma} f\left(\frac{x_i - \tau}{\sigma}\right) d\tau = \int_{(x_i - t)/\sigma}^{\infty} f(u) du = 1 - F\left(\frac{x_i - t}{\sigma}\right). \quad \square$$

6.2 Proof of Proposition 1.

Proof of Proposition 1. Fix a cutoff strategy profile with cutoff $\hat{x}$: citizen i participates if and only if $x_i < \hat{x}$.

Step 1: participation and the induced success threshold. At true threshold τ , the mass of participants is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = \Pr(\tau + \sigma \varepsilon_i < \hat{x}) = \Pr\left(\varepsilon_i < \frac{\hat{x} - \tau}{\sigma}\right) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ , with $N^* \rightarrow 1$ as $\tau \rightarrow -\infty$ and $N^* \rightarrow 0$ as $\tau \rightarrow \infty$. The protest succeeds when $N^*(\tau) \geq \tau$. The left side of this inequality is strictly decreasing in τ and the right side is strictly increasing, so there is exactly one crossing $\hat{\tau}$ solving $F((\hat{x} - \hat{\tau})/\sigma) = \hat{\tau}$, and success occurs exactly on $\tau \leq \hat{\tau}$.

Step 2: the marginal citizen. A citizen with signal x_i gains $\Pr(\tau \leq \hat{\tau} \mid x_i) d_p - \gamma_r c_r$ from participating. By Lemma 1 evaluated at $t = \hat{\tau}$ and $x_i = \hat{x}$,

$$\Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) = 1 - F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = 1 - \hat{\tau},$$

where the second equality substitutes the crossing condition from Step 1. Indifference of the marginal citizen requires $(1 - \hat{\tau}) d_p - \gamma_r c_r = 0$; solving,

$$1 - \hat{\tau} = \frac{\gamma_r c_r}{d_p} \iff \hat{\tau} = 1 - \frac{\gamma_r c_r}{d_p}.$$

Step 3: the cutoff signal. From the crossing condition, $F((\hat{x} - \hat{\tau})/\sigma) = \hat{\tau}$; since F is strictly increasing with full support, $(\hat{x} - \hat{\tau})/\sigma = F^{-1}(\hat{\tau})$, hence $\hat{x} = \hat{\tau} + \sigma F^{-1}(\hat{\tau})$.

Step 4: uniqueness within monotone strategies. It remains to show that only one cutoff is consistent with equilibrium. Differentiate the crossing condition implicitly with respect to $\hat{x}$, writing $z = (\hat{x} - \hat{\tau}(\hat{x}))/\sigma$:

$$\frac{f(z)}{\sigma} (1 - \hat{\tau}'(\hat{x})) = \hat{\tau}'(\hat{x}) \implies \hat{\tau}'(\hat{x}) = \frac{f(z)/\sigma}{1 + f(z)/\sigma} \in (0, 1).$$

The marginal citizen's gain at cutoff $\hat{x}$ is $(1 - \hat{\tau}(\hat{x}))d_p - \gamma_r c_r$, which is strictly decreasing in $\hat{x}$ because $\hat{\tau}'(\hat{x}) > 0$. Hence at most one cutoff makes the marginal citizen indifferent, and the pair $(\hat{x}, \hat{\tau})$ from Steps 2 and 3 is the unique monotone equilibrium. The extension of uniqueness beyond monotone strategies follows from iterated deletion of dominated strategies, as in Morris and Shin (2003); see also the discussion in Bueno de Mesquita and Shadmehr (2023).

Step 5: boundary policies. If $\gamma_r c_r \geq d_p$, the participation gain satisfies $\Pr(\cdot)d_p - \gamma_r c_r \leq d_p - \gamma_r c_r \leq 0$ for every belief, with strict inequality whenever the belief is interior, so no citizen participates and $N^* = 0 < \tau$ for every $\tau > 0$. If $c_r = 0$, the gain is $\Pr(\cdot)d_p \geq 0$, participation is weakly dominant, every citizen participates, and the protest succeeds for every $\tau \leq 1$, that is, $\hat{\tau} = 1$. $\square$

Remark 1. A natural question about the proof above is why the indifference condition is solved leaving $\hat{\tau}$ as is, instead of solving for the fixed point of the belief consistency condition. The answer is that both conditions are used, and the fixed point is indeed solved; it simply delivers the unknown that the analysis does not display. Note that the equilibrium is a system of two equations in two unknowns, $(\hat{x}, \hat{\tau})$. The belief consistency condition alone cannot be solved as a standalone fixed point, given that it is one equation in two unknowns. In other words, for every candidate cutoff $\hat{x}$ it defines an induced threshold $\hat{\tau}(\hat{x})$, and the indifference condition provides the second relation; the equilibrium is the intersection of the two.

The reason the algebra collapses is that both conditions depend on $(\hat{x}, \hat{\tau})$ only through the statistic $z = (\hat{x} - \hat{\tau})/\sigma$. Under the improper uniform prior, the posterior probability that the protest succeeds is $\Pr(\tau \leq t \mid x_i = \hat{x}) = 1 - F((\hat{x} - t)/\sigma)$ for any evaluation point t . Hence, evaluating at $t = \hat{\tau}$ and substituting $F(z) = \hat{\tau}$ is not circular; it is the ordinary elimination of one unknown using the other equation. The system becomes triangular. First, the two conditions jointly yield $(1 - \hat{\tau})d_p = \gamma_r c_r$, which pins down $\hat{\tau} = 1 - \gamma_r c_r/d_p$. Then, the belief consistency condition, used again in the other direction, recovers the cutoff signal $\hat{x} = \hat{\tau} + \sigma F^{-1}(\hat{\tau})$, which is the fixed point in question. The main text does not display it because the state's problem only requires $\hat{\tau}$, and in the small-noise limit $\hat{x} \rightarrow \hat{\tau}$, so the distinction vanishes where the state's stage operates.

Finally, note that this triangular structure is a consequence of the improper prior. Under a proper prior, the posterior does not reduce to $1 - F(z)$, the two conditions must be solved as a

genuine two-dimensional fixed point, and no closed form obtains.⁶

6.3 Proof of Proposition 2.

Proof of Proposition 2. Substituting $\hat{\tau}(c_r) = 1 - c_r/d_p$ into the objective and expanding term by term,

$$u_S(c_r) = (1 - \hat{\tau})B_s - \theta_0\hat{\tau} - \theta_1c_r^2 = \frac{c_r}{d_p}B_s - \theta_0\left(1 - \frac{c_r}{d_p}\right) - \theta_1c_r^2 = \frac{B_s + \theta_0}{d_p}c_r - \theta_0 - \theta_1c_r^2.$$

Differentiating twice, $u'_S(c_r) = (B_s + \theta_0)/d_p - 2\theta_1c_r$ and $u''_S(c_r) = -2\theta_1 < 0$: the objective is strictly concave, so the first order condition characterizes the unique maximizer on any interval. Setting $u'_S = 0$ gives $c_r^\circ = (B_s + \theta_0)/(2\theta_1d_p)$. If $c_r^\circ \leq d_p$, equivalently $B_s + \theta_0 \leq 2\theta_1d_p^2$, then $c_r^* = c_r^\circ$; otherwise $u'_S > 0$ on all of $[0, d_p]$ and the maximum over the interval is at d_p . For $c_r > d_p$ the threshold is fixed at $\hat{\tau} = 0$, so $u_S(c_r) = B_s - \theta_1c_r^2$, which is strictly decreasing; severity above d_p is therefore never optimal, and $c_r^* = \min\{c_r^\circ, d_p\}$. Finally, $u'_S(0) = (B_s + \theta_0)/d_p > 0$ by the parameter restrictions, so $c_r^* > 0$. $\square$

6.4 Proof of Proposition 3.

Proof of Proposition 3. At the interior solution $c_r^* = (B_s + \theta_0)/(2\theta_1d_p)$, differentiate with respect to each parameter in turn:

$$\frac{\partial c_r^*}{\partial B_s} = \frac{\partial}{\partial B_s} \left[\frac{B_s + \theta_0}{2\theta_1d_p} \right] = \frac{1}{2\theta_1d_p} > 0, \quad \frac{\partial c_r^*}{\partial \theta_0} = \frac{1}{2\theta_1d_p} > 0,$$

since the parameter enters the numerator linearly with unit coefficient in both cases;

$$\frac{\partial c_r^*}{\partial \theta_1} = \frac{B_s + \theta_0}{2d_p} \cdot \frac{\partial}{\partial \theta_1} [\theta_1^{-1}] = \frac{B_s + \theta_0}{2d_p} \cdot (-\theta_1^{-2}) = -\frac{B_s + \theta_0}{2\theta_1^2d_p} < 0,$$

$$\frac{\partial c_r^*}{\partial d_p} = \frac{B_s + \theta_0}{2\theta_1} \cdot \frac{\partial}{\partial d_p} [d_p^{-1}] = \frac{B_s + \theta_0}{2\theta_1} \cdot (-d_p^{-2}) = -\frac{B_s + \theta_0}{2\theta_1d_p^2} < 0;$$

⁶This is precisely why the improper prior is the standard device in this literature (Morris and Shin 2003; Bueno de Mesquita and Shadmehr 2023). At the critical threshold, the marginal citizen's belief about her rank in the population is uniform, so her success probability equals one minus the critical mass.

and B_p does not appear in the expression, so $\partial c_r^*/\partial B_p = 0$. For the success probability, substitute the interior solution into the threshold:

$$\hat{\tau}^* = 1 - \frac{c_r^*}{d_p} = 1 - \frac{B_s + \theta_0}{2\theta_1 d_p^2},$$

whence

$$\frac{\partial \hat{\tau}^*}{\partial B_s} = \frac{\partial \hat{\tau}^*}{\partial \theta_0} = -\frac{1}{2\theta_1 d_p^2} < 0, \quad \frac{\partial \hat{\tau}^*}{\partial \theta_1} = \frac{B_s + \theta_0}{2\theta_1^2 d_p^2} > 0, \quad \frac{\partial \hat{\tau}^*}{\partial d_p} = \frac{B_s + \theta_0}{\theta_1 d_p^3} > 0,$$

where the last derivative uses $\partial(d_p^{-2})/\partial d_p = -2d_p^{-3}$, so that $-\frac{B_s + \theta_0}{2\theta_1} \cdot (-2d_p^{-3}) = \frac{B_s + \theta_0}{\theta_1 d_p^3}$. B_p is again absent, so $\partial \hat{\tau}^*/\partial B_p = 0$. $\square$

6.5 Proof of Proposition 4.

Proof of Proposition 4. With $d_p(c_r) = d_0 + e(c_r)$ and $\gamma_r = 1$, apply the quotient rule to $\hat{\tau}(c_r) = 1 - c_r/d_p(c_r)$:

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_p(c_r) \cdot 1 - c_r e'(c_r)}{d_p(c_r)^2} = -\frac{d_p(c_r) - c_r e'(c_r)}{d_p(c_r)^2}.$$

The denominator is positive, so the derivative has the sign of $-(d_p(c_r) - c_r e'(c_r))$. Dividing the parenthesis by $d_p(c_r) > 0$,

$$d_p(c_r) - c_r e'(c_r) \geq 0 \iff 1 - \frac{c_r e'(c_r)}{d_p(c_r)} = 1 - \varepsilon(c_r) \geq 0,$$

so $\partial \hat{\tau}/\partial c_r < 0$ exactly when $\varepsilon(c_r) < 1$ and $\partial \hat{\tau}/\partial c_r > 0$ exactly when $\varepsilon(c_r) > 1$. $\square$

6.6 Proof of Proposition 5.

Proof of Proposition 5. Part (i). With $e(c_r) = \alpha c_r$, $e'(c_r) = \alpha$ and $d_p(c_r) = d_0 + \alpha c_r$, so

$$\varepsilon(c_r) = \frac{\alpha c_r}{d_0 + \alpha c_r} < 1 \iff \alpha c_r < d_0 + \alpha c_r \iff 0 < d_0,$$

which holds by assumption; by Proposition 4, severity always deters. For the derivative, the quotient-rule numerator is $(d_0 + \alpha c_r) - c_r \alpha = d_0$, hence $\partial \hat{\tau}/\partial c_r = -d_0/(d_0 + \alpha c_r)^2$.

Part (ii). Dividing numerator and denominator by c_r ,

$$\hat{\tau}(c_r) = 1 - \frac{c_r}{d_0 + \alpha c_r} = 1 - \frac{1}{d_0/c_r + \alpha} \rightarrow 1 - \frac{1}{\alpha} \quad \text{as } c_r \rightarrow \infty,$$

and the convergence is monotone from above by part (i), so $\hat{\tau}(c_r) > 1 - 1/\alpha$ at every finite severity. Full deterrence requires $\hat{\tau} = 0$, that is, $c_r = d_0 + \alpha c_r$, that is, $c_r(1 - \alpha) = d_0$. For $\alpha \geq 1$ the left side is nonpositive for every $c_r > 0$ while the right side is positive, so no finite severity attains it; for $\alpha < 1$ the unique solution is $c_r = d_0/(1 - \alpha)$.

Part (iii). Using $1 - \hat{\tau} = c_r/(d_0 + \alpha c_r)$, the objective is $u_S(c_r) = (B_s + \theta_0)c_r/(d_0 + \alpha c_r) - \theta_0 - \theta_1 c_r^2$.

By the quotient rule,

$$\frac{d}{dc_r} \left[\frac{c_r}{d_0 + \alpha c_r} \right] = \frac{(d_0 + \alpha c_r) \cdot 1 - c_r \cdot \alpha}{(d_0 + \alpha c_r)^2} = \frac{d_0}{(d_0 + \alpha c_r)^2},$$

so $u'_S(c_r) = (B_s + \theta_0)d_0/(d_0 + \alpha c_r)^2 - 2\theta_1 c_r$; differentiating again, using $d[(d_0 + \alpha c_r)^{-2}]/dc_r = -2\alpha(d_0 + \alpha c_r)^{-3}$,

$$u''_S(c_r) = -\frac{2\alpha(B_s + \theta_0)d_0}{(d_0 + \alpha c_r)^3} - 2\theta_1 < 0,$$

so the objective is strictly concave. Moreover, $u'_S(0) = (B_s + \theta_0)d_0/d_0^2 = (B_s + \theta_0)/d_0 > 0$ and u'_S is strictly decreasing and diverges to $-\infty$, so u'_S has exactly one positive root, characterized by rearranging $u'_S = 0$ into $(B_s + \theta_0)d_0 = 2\theta_1 c_r(d_0 + \alpha c_r)^2$. For $\alpha < 1$ the relevant domain is $[0, \bar{c}]$ with $\bar{c} = d_0/(1 - \alpha)$ the full-deterrence corner from part (ii); by concavity, the corner is optimal exactly when $u'_S(\bar{c}) \geq 0$. Substituting $\bar{c}$: first,

$$d_0 + \alpha \bar{c} = d_0 + \frac{\alpha d_0}{1 - \alpha} = d_0 \frac{(1 - \alpha) + \alpha}{1 - \alpha} = \frac{d_0}{1 - \alpha},$$

hence

$$u'_S(\bar{c}) = (B_s + \theta_0)d_0 \cdot \frac{(1 - \alpha)^2}{d_0^2} - 2\theta_1 \cdot \frac{d_0}{1 - \alpha} = \frac{(B_s + \theta_0)(1 - \alpha)^2}{d_0} - \frac{2\theta_1 d_0}{1 - \alpha}.$$

Multiplying through by $d_0(1 - \alpha) > 0$, the condition $u'_S(\bar{c}) \geq 0$ is equivalent to $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$, as stated; the interior case is its strict complement. For $\alpha \geq 1$ there is no corner and the interior root is always the optimum; consistently, $(1 - \alpha)^3 \leq 0$ and the corner condition can never hold.

Part (iv). Write the first order condition as $G(c_r, \alpha) = (B_s + \theta_0)d_0(d_0 + \alpha c_r)^{-2} - 2\theta_1 c_r = 0$. The partial derivatives are

$$\frac{\partial G}{\partial \alpha} = (B_s + \theta_0)d_0 \cdot (-2)(d_0 + \alpha c_r)^{-3} \cdot c_r = -\frac{2(B_s + \theta_0)d_0 c_r}{(d_0 + \alpha c_r)^3} < 0, \quad \frac{\partial G}{\partial c_r} = u_S''(c_r) < 0,$$

so by the implicit function theorem $\partial c_r^* / \partial \alpha = -G_\alpha / G_{c_r} < 0$: a leading minus applied to the ratio of two negatives. For the threshold, $\hat{\tau}(c_r, \alpha) = 1 - c_r(d_0 + \alpha c_r)^{-1}$ depends on α both directly and through c_r^* :

$$\left. \frac{\partial \hat{\tau}}{\partial \alpha} \right|_{c_r} = -c_r \cdot (-1)(d_0 + \alpha c_r)^{-2} \cdot c_r = \frac{c_r^2}{(d_0 + \alpha c_r)^2} > 0, \quad \frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_0}{(d_0 + \alpha c_r)^2} < 0,$$

hence

$$\frac{d \hat{\tau}^*}{d \alpha} = \frac{\partial \hat{\tau}}{\partial \alpha} + \frac{\partial \hat{\tau}}{\partial c_r} \cdot \frac{\partial c_r^*}{\partial \alpha} > 0,$$

the first term positive and the second the product of two negatives. □

6.7 Proof of Proposition 6.

Proof of Proposition 6. Part (i). With $e(c_r) = \alpha c_r^k$, $e'(c_r) = \alpha k c_r^{k-1}$, and the numerator of $-\partial \hat{\tau} / \partial c_r$ from the proof of Proposition 4 is

$$d_p(c_r) - c_r e'(c_r) = d_0 + \alpha c_r^k - c_r \cdot \alpha k c_r^{k-1} = d_0 - \alpha(k-1)c_r^k,$$

which is strictly decreasing in c_r (as $k > 1$), positive at $c_r = 0$, and negative for large c_r . Its unique zero solves $\alpha(k-1)\tilde{c}^k = d_0$, that is, $\tilde{c} = (d_0/(\alpha(k-1)))^{1/k}$. Hence $\hat{\tau}$ is strictly decreasing on $(0, \tilde{c})$ and strictly increasing on $(\tilde{c}, \infty)$: U-shaped with its minimum at $\tilde{c}$. For the value at the minimum, use $\alpha \tilde{c}^k = d_0/(k-1)$ to compute the denominator,

$$d_0 + \alpha \tilde{c}^k = d_0 + \frac{d_0}{k-1} = d_0 \frac{(k-1)+1}{k-1} = \frac{d_0 k}{k-1},$$

so that

$$\hat{\tau}(\tilde{c}) = 1 - \frac{\tilde{c}}{d_0 k / (k-1)} = 1 - \frac{\tilde{c}(k-1)}{d_0 k}.$$

Part (ii). Write the objective as $u_S(c_r) = (B_s + \theta_0)(1 - \hat{\tau}(c_r)) - \theta_0 - \theta_1 c_r^2$. On $(\tilde{c}, \infty)$, part (i) gives $\hat{\tau}$ strictly increasing, so $1 - \hat{\tau}$ is strictly decreasing and the first term falls, while $-\theta_1 c_r^2$ also falls; hence u_S is strictly decreasing beyond $\tilde{c}$ and no $c_r > \tilde{c}$ can be optimal: $c_r^* \leq \tilde{c}$.

Part (iii). Full deterrence requires $\hat{\tau}(c_r) \leq 0$ at some severity. By part (i) the minimum of $\hat{\tau}$ over all severities is $\hat{\tau}(\tilde{c})$; if $\hat{\tau}(\tilde{c}) > 0$, then $\hat{\tau}(c_r) > 0$ at every severity and no policy attains full deterrence. □

All of the claims in this appendix were additionally verified numerically over randomized parameter draws.